Innovation Centre for Education

YENEPOYA (DEEMED TO BE UNIVERSITY)

Project - High Level Design

on

Predictive Modeling for RetaiSuccess

Submitted by

Muhammed Shahabaz T M

23BCDSBA67

Industry Mentor:
Mr.Ujjwal Kumar

Guided by:
MS.Raksha Adyanthaya

TABLE OF CONTENTS

S.No	Content	Page No
1	Introduction	3
1.1	Scope of the Document	3
1.2	Intended Audience	3
1.3	System Overview	4
2	System Design	5
2.1	Application Design	5
2.2	Process Flow	6
2.3	Information Flow	7
2.4	Components Design	7
2.5	Key Design Considerations	8
2.6	API Catalogue	8
3	Data Design	9
3.1	Data Model	9
3.2	Data Access Mechanism	9
3.3	Data Retention Policies	10
3.4	Data Migration	10
4	Interfaces	11
5	State and Session Management	12
6	Caching	13
7	Non-Functional Requirements	14
7.1	Security Aspects	14
7.2	Performance Aspects	14
8	References	15

1. INTRODUCTION

The project titled “Predictive Modeling for Retail Success” aims to analyze retail data and predict future sales using data analytics and machine learning techniques.
This system helps businesses understand customer behavior, improve decision-making, and increase profitability by providing accurate sales predictions.

1.1 Scope of the Document

This document describes the overall design and structure of the system. It includes system architecture, components, data flow, and key functionalities.
The scope covers how the system processes retail data and generates predictions to support business operations.

1.2 Intended Audience

This document is intended for project guides, developers, evaluators, and stakeholders who are involved in the development and assessment of the project.
It helps them understand the system design and working of the project.

1.3 System Overview

The system is designed to collect retail data such as sales, products, and customer information.
It processes the data using analytical and machine learning techniques to generate predictions.
The output is displayed through a user-friendly interface, helping users make informed business decisions.

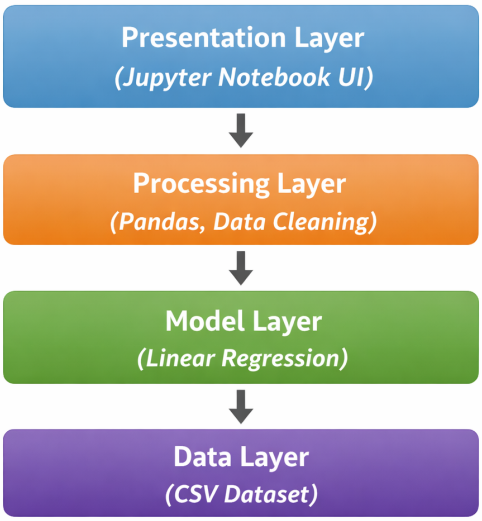
2. SYSTEM DESIGN

The system design defines the overall architecture of the application. It explains how different components interact with each other to process data and generate predictions. The system follows a modular approach, ensuring scalability, performance, and easy maintenance.

2.1 Application Design

The application is designed using a three-tier architecture consisting of the user interface, backend processing, and database layer.

The user interface allows users to input data and view results. The backend processes the data and applies prediction algorithms, while the database stores all relevant information.

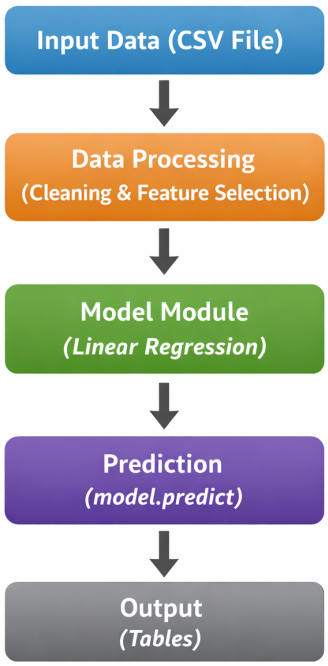


2.2 Process Flow

The system follows a structured flow:

Input Data → Data Preprocessing → Model Training → Prediction → Output Display

First, raw data is collected and cleaned. Then, it is processed using machine learning algorithms to generate predictions, which are displayed to the user.



2.3 Information Flow

Information flows from the user to the system through the input interface. The data is then processed in the backend and stored in the database.

After processing, the results are sent back to the user interface for display.

2.4 Components Design

The system consists of the following main components:

- Data Collection Module – Collects input data from users or datasets
- Data Processing Module – Cleans and prepares data
- Prediction Module – Applies machine learning algorithms
- User Interface Module – Displays output to users

2.5 Key Design Considerations

The system is designed considering the following factors:

- Accuracy – Ensures reliable prediction results
- Performance – Fast processing of data
- Scalability – Ability to handle large datasets
- Usability – Easy-to-use interface

2.6 API Catalogue

APIs are used to enable communication between the frontend and backend. These APIs handle data input, processing requests, and output generation.

Example:

- Input API → Receives user data
- Processing API → Sends data for prediction
- Output API → Returns prediction results

3. DATA DESIGN

The data design defines how data is structured, stored, and managed within the system. It includes details about data models, storage mechanisms, and policies for handling data efficiently and securely.

3.1 Data Model

The system uses a structured data model to organize retail data into different tables. The main entities include customers, products, and sales. Each entity contains specific attributes required for processing and prediction.

Example tables:

- Customer (Customer_ID, Name, Age, Location)
- Product (Product_ID, Name, Category, Price)
- Sales (Sale_ID, Product_ID, Quantity, Date)

3.2 Data Access Mechanism

Data is accessed using database queries and retrieval methods.
The system uses structured query language (SQL) to insert, update, and fetch data from the database.
Proper indexing is used to improve performance and speed of data access.

3.3 Data Retention Policies

The system maintains data for analysis and future reference.
Important data such as sales and customer records are stored securely for a defined period.
Old or unnecessary data may be removed or archived to optimize storage and maintain system performance.

3.4 Data Migration

Data migration refers to transferring existing data into the system.
The system supports importing data from external sources such as CSV or Excel files.
Proper validation is performed to ensure data accuracy during migration.

4. INTERFACES

Jupyter Notebook Interface

* Provides an interactive environment
* Allows step-by-step execution
* Displays outputs such as tables and graphs

User Interaction Interface

* User provides dataset or input values
* System processes data and returns predictions
* Results are shown in a simple and understandable format

5. STATE AND SESSION MANAGMENT

* The system is stateless in nature for retail prediction tasks
* Each execution processes the retail dataset independently
* No dependency on previous sales data runs or sessions
* No user login or authentication system required
* Retail data (sales, product, customer details) is:
    * Loaded at runtime
    * Processed temporarily in memory
* Machine learning model is:
    * Trained during execution
    * Not stored permanently unless saved manually
* All processed data and model results are cleared after execution
* Suitable for:
    * Offline retail data analysis
    * Single-user environment (Jupyter Notebook)
* Reduces system complexity and avoids session/database overhead
* Limitation:
    * Cannot track user activity or previous predictions
* Future enhancement:
    * Add session management for:
        * Multi-user retail systems
        * Web-based dashboards
        * Saving past prediction results

6. CACHING

Caching is a technique used to store frequently accessed data to improve system performance. In this predictive modeling system, caching is not used because the application is simple and processes a small dataset. The system runs in a Jupyter Notebook environment, where data is loaded and processed from the beginning during each execution. Since the processing time is minimal, caching is not required. This approach keeps the system simple, ensures accurate results, and avoids the use of additional memory.

Key POINTS:

* No caching mechanism is implemented
* Data is processed freshly in each run
* Dataset size is small, so processing is fast
* Avoids complexity in system design
* Ensures up-to-date and accurate results
* Reduces memory usage

7. NON-FUNCTIONAL REQUIRMENTS

Non-functional requirements define the quality attributes of the system such as performance, security, and reliability. These requirements ensure that the system operates efficiently and provides a good user experience.

In this project, the predictive modeling system for retail success is designed to efficiently process retail data and generate accurate sales predictions while maintaining data security and performance.

7.1 Security Aspects

The system ensures that data is handled securely and protected from unauthorized access. Basic security measures such as data validation and controlled access are implemented to maintain data integrity. Sensitive information is stored safely, and proper precautions are taken to prevent data misuse or loss.
Since the system uses a local dataset stored in a CSV file, the data is not shared externally. This ensures basic data privacy and reduces the risk of data breaches.

7.2 Performance Aspects

The system is designed to provide fast and efficient processing of data. It can handle datasets without significant delay and generate predictions quickly.

The use of lightweight libraries such as Pandas and Scikit-learn ensures faster execution and efficient computation. The system runs smoothly on standard hardware and provides quick results, making it suitable for real-time analysis in a simple environment.

8. REFERENCES

The following resources were referred to during the development of this project:

- Study materials and notes related to data analysis and machine learning
- Online tutorials and documentation for Python and data processing
- Research articles and websites related to predictive modeling
- Official documentation of tools and technologies used in the project

These references helped in understanding the concepts and implementing the system effectively.